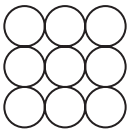
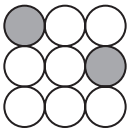
20 Dan has a grid of nine circles.



Dan chooses at random two of the circles in the grid.

(a) Show that the probability that Dan chooses the two circles shown shaded below is $\frac{1}{36}$



Either order: $\frac{1}{9} \times \frac{1}{8} + \frac{1}{9} \times \frac{1}{8}$

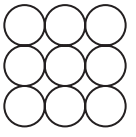
1 mark
Probability of one order.

$= 2 \times \frac{1}{72} = \frac{1}{36}$

1 mark
Two possible orders.

(2)

Joan also has a grid of nine circles.



Joan chooses at random three of the circles in the grid.

(b) Find the probability that these three circles are in a straight line.

Straight lines: 3 rows + 3 columns + 2 diagonals = 8.

1 mark
8 winning lines.

Ordered triples: $9 \times 8 \times 7 = 504$.

Each triple has $3 \times 2 \times 1 = 6$ possible orders.

1 mark
84 equally likely triples.

So distinct triples = $504 \div 6 = 84$.

Probability = $\frac{8}{84} = \frac{2}{21}$

1 mark
2/21.

(3)

(Total for Question 20 is 5 marks)

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